

Utilization of environmental and epidemiological indicators in the study of malaria dynamics

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ABSTRACT

This paper aims to analyze the behavior of malaria transmission in the Amazon region based on climatic and environmental changes, such as temperature, precipitation, deforestation and forest fires, through proposed modifications to the SIR and SEI models, in order to contribute to the study of applications of external effects on the evolution of the disease. The Trajetórias Project, developed by the Synthesis Center on Biodiversity and Ecosystem Services (SinBiose/CNPq) was used as an initial reference for the study.

This work employs a modified SEIRS/SEI methodology, based on the SIR/SEI model developed by Parham and Michael (2010) which takes into account rainfall and temperature, with further modifications.

1. Introduction

Malaria remains one of the most significant vector-borne diseases worldwide, caused by protozoa of the genus *Plasmodium* and transmitted by female mosquitoes of the genus *Anopheles*. Although present in multiple regions, the burden of malaria in the Americas is highly concentrated in the Amazon basin, which spans nine countries and accounts for a substantial proportion of reported cases [1]. The persistence of malaria in this region is closely linked to ecological, climatic, and socioeconomic factors that influence both vector dynamics and human exposure.

Environmental changes play a central role in shaping malaria transmission. In the Amazon, processes such as deforestation, agricultural expansion, and infrastructure development alter local ecosystems, affecting vector habitats and increasing human–vector contact. These changes often promote the proliferation of mosquito breeding sites and concentrate human populations in high-risk areas, particularly among non-immune migrants [4, 5]. In addition to land-use change, climatic variables such as temperature and rainfall, as well as disturbances such as forest fires and mining activities, further influence vector abundance, pathogen development, and transmission intensity.

Despite the recognized importance of environmental drivers, integrating these factors into mathematical models of malaria transmission remains challenging. In particular, distinguishing between their effects on mosquito biology and their impact on human–vector interaction is essential for developing models that are both biologically consistent and identifiable from available data.

In this work, we develop and analyze a modified SIR–SEI

model for malaria transmission in the Amazon, incorporating environmental variables such as temperature, rainfall, deforestation, and fire activity. Climatic factors are modeled as drivers of mosquito life-history parameters, while land-use variables are introduced as modifiers of transmission intensity. Using epidemiological and environmental data from the Trajetorias Project [6], we investigate how these factors influence disease dynamics and assess their relative contributions through analytical and numerical methods.

2. Model formulation

2.1. Base SIR–SEI model

Based on previous work developed to model the transmission of malaria based on precipitation and temperature dynamics [7], we initially focused on two different sets of compartments of hosts and vectors: susceptible, infected and recovered hosts (S_H , I_H , R_H respectively), and susceptible, exposed and infected vectors (S_M , E_M , I_M respectively).

The equations that describe the transmission are initially given as:

$$\frac{dS_H}{dt} = -ab_2 \left(\frac{I_M}{N} \right) S_H \quad (1)$$

$$\frac{dI_H}{dt} = ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) - \gamma I_H \quad (2)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (3)$$

$$\frac{dS_M}{dt} = b - \left(ab_1 \left(\frac{I_H}{N} \right) + \mu \right) S_M \quad (4)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \left(\frac{I_H}{N} \right) S_M - \mu E_M \\ & - ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \mathcal{L}(\tau_M) \end{aligned} \quad (5)$$

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$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell(\tau_M) \\ & - \mu I_M \end{aligned} \quad (6)$$

The parameters are given in **Table A.1**, while the variables are given in **Table A.2**, which can be found in the Appendix.

However, the use of these equations proved to be unsuccessful in the appropriate modelling of the behavior of the disease, as can be seen in Figures 1A and 2A: while it would be expected for the number of infected humans to oscillate during the period of study, given that malaria is a recurring illness in the region, the model indicates that the number of infections would rapidly fall to 0, without occurrence of new cases.

In addition to that, it could also be seen that the mosquito population would also decline to near negligible levels, which could explain the lack of infections in the human population.

To overcome this effect, the SEI equations were modified in order for b_1 to only be used in the transference between compartments S and E , and a new parameter, b_3 to move mosquitos from E to I . Being inversely related to the incubation period, the definition of this rate can be found in **Table A.2**. In addition to that, the parameter $a(T)$ that was in use in the transition between exposed and infected also had to be removed, as no further bites occur at this point. The parameter l also had to be disassociated from the infection rate of exposed individual (b_3).

The SIR equations were also modified, to take into consideration population dynamics, including a r_H parameter as the net human increase rate per capita, which could be explained by births or immigration to the municipality. Given the positive growth of the studied population, a death rate was neglected for this study.

The updated equations used in the model are as following:

$$\frac{dS_H}{dt} = r_H N - ab_2 \left(\frac{I_M}{N} \right) S_H \quad (7)$$

$$\frac{dI_H}{dt} = ab_2 \left(\frac{I_M}{N} \right) S_H - \gamma I_H \quad (8)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (9)$$

$$\frac{dS_M}{dt} = b - \left(ab_1 \left(\frac{I_H}{N} \right) + \mu \right) S_M \quad (10)$$

$$\frac{dE_M}{dt} = ab_1 \left(\frac{I_H}{N} \right) S_M - (\mu + b_3 + l) E_M \quad (11)$$

$$\frac{dI_M}{dt} = b_3 E_M - \mu I_M \quad (12)$$

2.2. Base SEIRS-SEI model

Preliminary analysis of the baseline SIR-SEI framework revealed structural limitations in replicating endemic malaria transmission, as the system intrinsically tended toward a single outbreak followed by rapid fade-out. To rectify this and enhance biological fidelity, we extended the model to a SEIRS-SEI structure. This extension accounts for essential features of Plasmodium dynamics, such as the extrinsic incubation period in humans, represented by the Exposed compartment (E_H), and the decay of temporary acquired immunity, represented by the transition from Recovered to Susceptible ($R_H \rightarrow S_H$). These modifications are necessary to sustain the multi-year recurrence and seasonal oscillations characteristic of endemic regions.

The constructed model is as follows:

$$\frac{dS_H}{dt} = r_H N - ab_2 \left(\frac{I_M(t)}{N} \right) S_H + \omega R_H \quad (13)$$

$$\begin{aligned} \frac{dE_H}{dt} = & ab_2 \left(\frac{I_M(t)}{N} \right) S_H \\ & - ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{dI_H}{dt} = & ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) \\ & - \gamma I_H \end{aligned} \quad (15)$$

$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \quad (16)$$

$$\frac{dS_M}{dt} = \Lambda - \left(ab_1 \left(\frac{I_H(t)}{N} \right) + \mu \right) S_M \quad (17)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \left(\frac{I_H(t)}{N} \right) S_M - \mu E_M \\ & - ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell \\ & - \mu I_M \end{aligned} \quad (19)$$

Where Λ is the mosquito recruitment rate:

$$\Lambda = b(R, T) \cdot K \cdot \left(1 - \frac{S_M + E_M + I_M}{K} \right) \quad (20)$$

And K is the mosquito carrying capacity:

$$K = M \times \text{Temp. factor} \times \text{Rain factor} \times \text{Seasonal factor} \quad (21)$$

Where:

$$\text{Temp. factor} = \max \left(0.1, \min \left(1, \frac{T - 15}{15} \right) \right) \quad (22)$$

$$\text{Rain factor} = \min \left(1, \max \left(0, \frac{R}{30} \right) \right) \quad (23)$$

$$\text{Seasonal factor} = 2.5 + 2 \sin \left(2\pi \frac{t - 180}{365.25} \right) \quad (1 \leq t \leq 366) \quad (24)$$

3. Data description and processing

It was decided to use data from cases caused by *P. vivax* as this species is responsible for the highest number of malaria cases in Brazil [11, 12], being the most common human malaria parasite in the region [13].

Based on higher incidence and number of cases, the municipality initially chosen for analysis was of Manaus, especially the rural zones, as it had the highest incidence of malaria caused by *P. vivax* in the Amazon region. Now, as it was not possible to identify the population in rural areas based on available census data, we used the Trajetórias Project to estimate this population based on the incidence function given [6]:

$$\text{Inc} = \frac{\text{Cases}(d, m, z, t_1, t_2)}{\text{Pop}(m, z, (t_1 + t_2)/2) \times 5 \text{ years}} \times 10^5, \quad (25)$$

where $\text{Cases}(d, m, z, t_1, t_2)$ is the number of cases of disease d in zone z of municipality m , and t_1 and t_2 are the initial and final years of the interval, while $\text{Pop}(m, z, (t_1 + t_2)/2) \times 5$ years is the population in zone z of municipality m in the middle of the period multiplied by the total number of observation years. In the case of the rural population, we could indicate it as:

$$\text{Inc}(\text{Viv.}, \text{Man.}, \text{Ru.}, 2004, 2008) = \quad (26)$$

$$= \frac{\text{Cases}(\text{Viv.}, \text{Man.}, \text{Rur.}, 2004, 2008)}{\text{Pop}(\text{Man.}, \text{Rur.}, 2006) \times 5 \text{ years}} \times 10^5 \quad (27)$$

$$\Rightarrow 184030.8 = \frac{78745}{5\text{Pop}} \times 10^5 \Rightarrow \text{Pop} \approx 8558 \quad (28)$$

Using data on the total population of Manaus from 2004 to 2008, with an incidence of 3106.429047 and a number of cases of 262,264, the total population of the municipality was estimated to be 1,688,524 inhabitants, which corresponds to the value found in the official census [14]. Thus, the rural population represented approximately 0.5% of the municipality's population in this period, while the number of cases in rural setting is about 30% of all cases of malaria caused by *P. vivax* in Manaus in this period, and about 9.4% of the cases in the state.

Using data from 2015 to 2019, the rural population is found to represent approximately 0.4% of the municipality's population, the number of cases in rural settings being about 41.5% of all cases in the city in this period, and about 7.2% of the cases in the state.

The yearly population was constructed via interpolation using data from IBGE (Brazilian Institute of Geography and Statistics, governmental body responsible for the official Brazilian censuses), based on the censuses of 2000, 2010 and 2022, and an estimation of the population of 2025 [15, 16, 17].

As the data on malaria cases from the Trajetórias project was not given annually, but instead as a sum of cases in each five-year period of study, yearly data was obtained from SIVEP [18], which registers notifications from every brazilian municipality from 2003 to 2023.

In order to use the latest and complete available data on registered cases and deforestation and forest fires noticed, this study was done using data from 2016 to 2024, but limited to the years of 2017 to 2023, given that the SIVEP database didn't include registered data from 2024 at the time of development of this study.

As to properly approximate the rural population of Manaus and the number of rural cases in the municipality, the rural proportion for 2020 to 2023 needed to be extrapolated from the estimated 2017 data, given that it was the midpoint of the latest estimate from Trajetórias.

During the preprocessing stage of data from the Trajetórias project, several inconsistencies were identified and corrected. First, the rural and urban incidence rates for the period 2015–2019 had been originally computed using population estimates from 2004–2008, leading to a temporal mismatch. These incidence values were recalculated using population data corresponding to the appropriate time period.

Additionally, the rural and urban population estimates for 2006 and 2017 had initially been derived using a fixed proportional split based on 2010 data. To better reflect demographic changes over time, particularly those associated with urbanization, these estimates were revised. The rural population for 2006 was approximated as the mean between the reported values for 2000 and 2010, while the 2017 population was obtained through extrapolation based on the observed growth trend between these years. This approach provides a more consistent and temporally coherent representation of population dynamics across the study period.

4. Impacts of environmental factors

Environmental changes, particularly deforestation and forest fires, have been widely associated with the resurgence and spatial redistribution of malaria transmission in the

Amazon region [20]. The removal of forest canopy alters local microclimatic conditions, increasing sunlight exposure and ambient temperature, which in turn favor the development of mosquito larvae and pupae. In particular, the primary malaria vector in the Amazon, *Anopheles darlingi*, is known to thrive in partially deforested environments, where sunlit water bodies promote larval growth and increase the availability of food sources such as algae [21, 22].

Beyond ecological effects, deforestation also influences human exposure to malaria. Activities such as logging, agriculture, and infrastructure development lead to increased human presence in recently altered environments, often in close proximity to mosquito breeding sites. This results in higher human–vector contact rates and facilitates transmission, especially in areas where both vector density and environmental suitability are elevated. Additionally, land-use changes such as agriculture may contribute to river sedimentation and the formation of artificial breeding sites, further enhancing mosquito proliferation.

Forest fragmentation has been shown to accelerate mosquito development cycles by increasing temperature and light availability, reducing the time from larval to adult stages and increasing vectorial capacity and survivorship [27, 28]. Similarly, forest fires, often used as a land-clearing mechanism, remove vegetation and modify habitat structure, contributing to long-term ecological changes that can influence mosquito populations and disease transmission dynamics [29, 30]. These processes may also increase the likelihood of subsequent fires by promoting understory drying and fuel accumulation [31].

In this context, deforestation and fire activity are incorporated into the model as external drivers that modify transmission dynamics. Rather than directly altering mosquito physiological parameters, these factors are interpreted as influencing human–vector contact rates and exposure, thereby affecting the effective transmission of the disease.

Deforestation data were obtained from the DETER system, which provides daily records of deforestation alerts by degradation type, reported in km² [34]. The data were aggregated to produce a complete daily time series of deforested area. Forest fire data were obtained from the Brazilian National Institute for Space Research (INPE), which provides daily records of detected fire hotspots [35]. Due to the absence of area-based measurements at this temporal resolution, the number of detected hotspots was used as a proxy for fire activity.

5. Mathematical analysis of the model

5.1. Basic Reproduction Number ($\mathcal{R}_0$)

To derive the $\mathcal{R}_0$ of the model, we define X_s as the set of all disease-free states,

$$X_s = \{x \geq 0 | x_i = 0, i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^T$, such that $x_i \geq 0$ represents the number of individuals in each compartment, and each function is assumed to be continuously differentiable at least twice in each variable (C^2) [36].

Rearranging the equations so that the first m contain the infected individuals, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i , $\mathcal{V}_i^-(x)$ be the rate of individuals entering compartment i by other means, and $\mathcal{V}_i^+(x)$ be the rate of individuals leaving compartment i . The disease transmission model consists of non-negative initial conditions together with the following system of equations:

$$\dot{x} = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), i = 1, \dots, n,$$

where $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. Here, $F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j} \right]$ and $V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j} \right]$, where x_0 is a Disease-Free Equilibrium (DFE), and $1 \leq i, j \leq m$, and $\mathcal{R}_0 = \rho(FV^{-1})$.

The formulation for $\mathcal{R}_0$ in both model, as well as the complete SIR/SEI system is given by:

SIR: In this case, $m = 1$, and the compartments are arranged as $[I_H, S_H, R_H]$. Since $\mathcal{R}_0$ is calculated with normalized values, the necessary equations are multiplied by N to remove the denominator. Specifically, for the SIR case, as R_H is not used in any of the equations, it can be expressed solely in terms of S and I . Therefore:

- $\mathcal{F}_i(x)$: rate of entrance of infected individuals in compartment i
- $\mathcal{V}_i(x)^-$: rate of leaving compartment i
- $\mathcal{V}_i(x)^+$: rate of entering compartment i

So we have

$$\begin{aligned} F &= [ab_2 I_M S_H] \\ \mathcal{V}^- &= [\gamma I_H] \\ \mathcal{V}^+ &= [0] \\ \mathcal{V}_i(x) &= \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+ \end{aligned}$$

Therefore,

$$\mathcal{V} = [I_H \gamma]$$

Hence

$$\begin{aligned} F &= \frac{\partial F}{\partial I_M} = [ab_2 S_H] \\ V &= \frac{\partial V}{\partial I_H} = [\gamma] \end{aligned}$$

At the equilibrium, $[S_H^*, I_H^*] = [1, 0]$, so

$$F = [ab_2], \quad V = [\gamma] \text{ and } \mathcal{R}_0 = \left| \frac{ab_2}{\gamma} \right|.$$

SEI: In this case, $m = 2$, and the compartments are arranged as $[E_M, I_M, S_M]$. Again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$\mathcal{F} = \begin{bmatrix} ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$\begin{aligned} F &= \frac{\partial \mathcal{F}}{\partial E_M, I_H} \\ &= \begin{bmatrix} \frac{\partial ab_1 I_H S_M}{\frac{\partial E_M}{\partial 0}} & \frac{\partial ab_1 I_H S_M}{\frac{\partial I_H}{\partial 0}} \\ \frac{\partial 0}{\partial E_M} & \frac{\partial 0}{\partial I_H} \end{bmatrix} \\ &= \begin{bmatrix} 0 & ab_1 S_M \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} V &= \frac{\partial \mathcal{V}}{\partial E_M, I_M} \\ &= \begin{bmatrix} \frac{\partial E_M(\mu + b_3 + l)}{\frac{\partial \mu I_M - b_3 E_M}{\partial E_M}} & \frac{\partial E_M(\mu + b_3 + l)}{\frac{\partial \mu I_M - b_3 E_M}{\partial I_M}} \\ \frac{\partial \mu I_M - b_3 E_M}{\partial E_M} & \frac{\partial \mu I_M - b_3 E_M}{\partial I_M} \end{bmatrix} \\ &= \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix} \end{aligned}$$

At the equilibrium, $[S_M^*, E_M^*, I_M^*] = [1, 0, 0]$, so

$$\begin{aligned} F &= \begin{bmatrix} 0 & ab_1 \\ 0 & 0 \end{bmatrix}, \\ V &= \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix} \end{aligned}$$

$$\text{and } \mathcal{R}_0 = \left| \frac{ab_1 b_3}{(b_3 + l + \mu)\mu} \right|.$$

SIR/SEI: In this case, $m = 3$, and the compartments are arranged as $[I_H, E_M, I_M, S_H, S_M]$. Once again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$\mathcal{F} = \begin{bmatrix} ab_2 I_M S_H \\ ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} \gamma I_H \\ E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} I_H \gamma \\ E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$F = \frac{\partial \mathcal{F}}{\partial I_H, E_M, I_M} = \begin{bmatrix} 0 & 0 & ab_2 S_H \\ ab_1 S_M & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$V = \frac{\partial \mathcal{V}}{\partial I_H, E_M, I_M} = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

At the equilibrium,

$$[S_H^*, S_M^*, I_H^*, E_M^*, I_M^*] = [1, 1, 0, 0, 0],$$

so

$$F = \begin{bmatrix} 0 & 0 & ab_2 \\ ab_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \left| \sqrt{\frac{a^2 b_1 b_2 b_3}{(b_3 + l + \mu)\gamma\mu}} \right| = \sqrt{\mathcal{R}_{0SIR} \times \mathcal{R}_{0SEI}}$$

5.2. Understanding relations

Having the necessary epidemiological and environmental datasets ready for use, the next step taken was to verify the correlation between the number of cases and forest areas impacted by deforestation and fires, in addition to the correlation with temperature and precipitation as well.

Using the Pandas library on a Python environment, the correlation between monthly number of cases caused by *P. vivax* in Manaus and deforested area was found to be approximately -0.11, and 0.137 between cases and burnt area. In relation to climatic data, the correlation with temperature was found to be around 0.404, and with precipitation to be -0.365. Both correlation tables were taken in relation to monthly data.

On the other hand, taking the correlation between the incidence of cases in relation to the seasons, deforested and burnt areas, that was found to be 0.722, 0.314 and 0.818, respectively. Looking for the correlation in relation to seasons, temperature and precipitation, that was found to be 0.687, 0.839 and -0.883, respectively.

6. Results

The initial implementation of the model using the original functional forms and parameter values reported in [7] did not reproduce biologically consistent transmission dynamics, particularly in the mosquito population. As illustrated in Figures 1–3A, the human SIR component exhibits a monotonic decline in the number of infected individuals, suggesting that the infection fails to persist in the population under the given parametrization. More critically, the corresponding SEI mosquito model yields nonphysical behavior, with the susceptible mosquito population attaining negative values and exhibiting oscillatory dynamics.

These inconsistencies arise from the functional forms adopted for key entomological parameters, which do not explicitly enforce biologically meaningful bounds. In particular, probabilities and rates are not constrained to lie within the interval $[0, 1]$, leading to violations of positivity and instability in the numerical solutions.

To address this issue, the functional forms were modified to ensure that all probabilities and rates remain within their admissible ranges. The resulting dynamics, shown in Figure 3A, exhibit qualitatively different behavior, with the mosquito population rapidly declining toward extinction. While this modification restores mathematical consistency, it also highlights a mismatch between the model structure and the expected persistence of malaria in endemic settings, motivating the need for further refinement of the transmission and demographic assumptions.

Before including the environmental factors on the used model (7-12), its results were compared to a model based on SIVEP data, keeping use of equations (7, 10-12), but using

$$I_H(t) = \sum_{k=\max\left(0, t-\frac{1}{\gamma}+1\right)}^{0 \leq k \leq 1825} \# \text{ of cases} \quad (29)$$

$$R_H(t) = \max\left(0, N(1 + r_H t) - S_H(t) - I_H(t)\right) \quad (30)$$

$I_H(t)$ now being the moving sum of cases over a period of 120 days, as it is the average duration of infectiousness, and $R_H(t)$ being the difference between the initial population and the susceptible and infected population at each day, with the addition of the daily births. The result can be seen in the graph below.

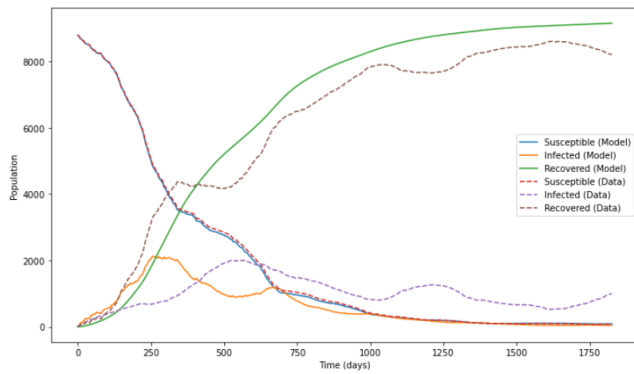


Figure 1: Comparison between observed data of infections and model estimation of I_H .

It is possible to see that the model gives a close approximation to the format of the observed number of infections and recoveries, although the infections appear to increase faster and die out given the model, a behavior that is not seen on the actual data.

At this point, a linear combination was implemented as a replacement to the use of simply T and R , in order to take into consideration the effects of deforestation and fires, with the α -function $\alpha_1 D + \alpha_2 F + \alpha_3 T + \alpha_4 R$, with $\alpha_0 = [0, 0, 1, 1]$ representing the primary version of the model.

7. Computational Implementation

All model simulations were performed using Python version 3.12.7 (conda-forge distribution). Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

The systems of time-delay ordinary differential equations (1-6) were solved numerically using the explicit Runge-Kutta method of order 8 as implemented in the `solve_ivp` function from SciPy version 1.17.1. All computations were performed on a laptop running **Windows 11** (64-bit) equipped with a **12th Gen Intel Core i5-1235U** processor (10 cores, 12 threads, 1.30 GHz base frequency) and **16 GB RAM**.

The source code and data required to reproduce all figures are available at https://github.com/RaphaLevy/Study_on_malaria_dynamics_v2.

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A. Appendix

A.1. Tables

Table A.1

List of model parameters.

Parameter	Definition	Formulation
$b(R, T)$	Mosquito birth rate (days ⁻¹)	$B_E p_E(R) p_L(R, T) p_P(R) / (\tau_E + \tau_L(T) + \tau_P)$
$a(T)$	Biting rate (days ⁻¹)	$(T - T') / D_1$
$\mu(T)$	Mosquito mortality rate per capita (days ⁻¹)	$-\log(p(T))$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD / (T - T_{min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1 / (c_1 T + c_2)$
$p(T)$	Daily mosquito survival rate	$e^{(-1 / (AT^2 + BT + C))}$
$p_L(R)$	Probability of larval survival dependent on rainfall	$(4p_{ML} / R_L^2) R(R_L - R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$e^{-(c_1 T + c_2)}$
$p_L(R, T)$	Probability of larval survival	$p_L(R) p_L(T)$
$l(\tau_M)(T)$	Probability of mosquito survival during sporozoite cycle (days ⁻¹)	$p(T)^{\tau_M(T)}$
$M(t)$	Total population of mosquitos	$S_M(t) + E_M(t) + I_M(t)$
$N(t)$	Total population of humans	$S_H(t) + I_H(t) + R_H(t)$

Table A.2

List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection	Dimensionless
γ	1/Average duration of infectiousness in humans	days ⁻¹
T_1	Mean temperature in the absence of seasonality	°C
T_2	Amplitude of seasonal variability in temperature	Dimensionless
ω_1	Angular frequency of seasonal oscillations in temperature	months ⁻¹
ϕ_1	Phase lag of temperature variability (phase shift)	Dimensionless
R_1	Average monthly precipitation in the absence of seasonality	mm
R_2	Amplitude of seasonal variability in precipitation	Dimensionless
ω_2	Angular frequency of seasonal oscillations in precipitation	months ⁻¹
ϕ_2	Phase lag of precipitation variability (phase shift)	Dimensionless
B_E	Number of eggs laid per adult per oviposition	Dimensionless
p_{ME}	Maximum probability of egg survival	Dimensionless
p_{ML}	Maximum probability of larval survival	Dimensionless
p_{MP}	Maximum probability of pupal survival	Dimensionless
τ_E	Duration of the egg development phase	days
τ_P	Duration of the pupal development phase	days
τ_H	Latent period of infection within humans	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 [8]	mm
T_{min}	Minimum temperature, below which there is no development of the parasite: 14.5	°C
DD	Degree-days for parasite development. "Sum of heat" for maturation: 105 [9, 10]	°C days
A	Empirical sensitivity parameter	(°C ² days) ⁻¹
B	Empirical sensitivity parameter	(°C days) ⁻¹
C	Empirical sensitivity parameter	days ⁻¹
D_1	Temperature scaling factor of the biting rate	°C days
c_1	Empirical sensitivity parameter	(°C days) ⁻¹
c_2	Empirical sensitivity parameter	days ⁻¹
T'	Biting rate temperature threshold ¹	°C
b_3	Infection rate in exposed mosquitoes ($1/\tau_M(T)$)	days ⁻¹
r_H	Net daily recruitment rate per capita	days ⁻¹

¹ In Parham and Michael [7], this parameter is denoted T_1 . We rename it here to avoid confusion with the mean temperature in the absence of seasonality, also denoted T_1 in the original paper.

A.2. Figures

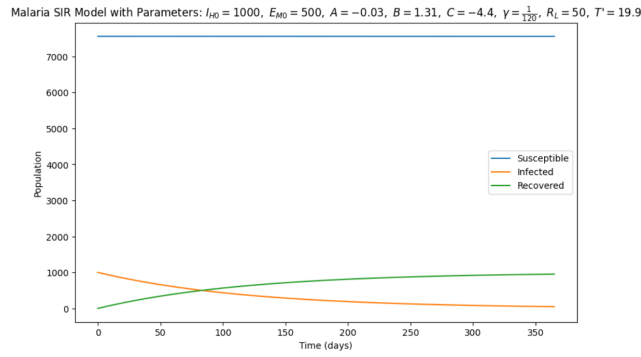


Figure 1A: Simulation of the baseline SIR model using the original parameterization from [7]. The dynamics show a monotonic decline in the number of infected individuals, indicating the absence of sustained transmission.

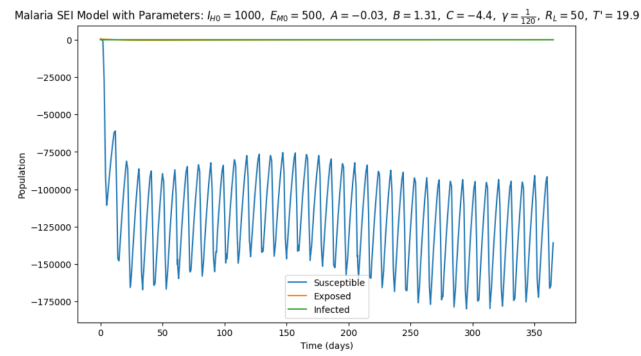


Figure 2A: Simulation of the baseline SEI mosquito model using the original parameterization from [7]. The susceptible mosquito population exhibits oscillatory behavior and attains negative values, indicating a loss of biological and mathematical consistency.

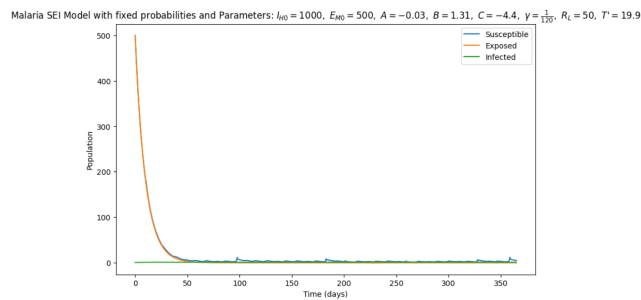


Figure 3A: Simulation of the SEI mosquito model after modifying parameter functions to enforce biologically admissible bounds on probabilities and rates. Although positivity is restored, the mosquito population rapidly declines toward extinction, suggesting that additional mechanisms are required to sustain endemic transmission.